

Explainable Integration of Process and Optical Parameters for Plasma Etch Depth Prediction

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Accurate prediction and understanding of plasma etch depth are crucial for process stability and yield in semiconductor manufacturing. While conventional machine learning approaches demonstrate the feasibility of non-contact prediction, they often lack physical interpretability and struggle with complex feature dependencies. In this study, we develop an explainable machine learning framework that integrates plasma process parameters and optical features from digital image colorimetry (DIC) to predict the remaining SiO₂ thickness after plasma etching. To account for limited experimental data and measurement variability, a probabilistic treatment was applied by introducing spatial/parametric variations and input noise. Multiple regression models were systematically compared; the ensemble-based Random Forest Model (RFM) demonstrated superior performance (RMSE < 1.4 nm) and greater robustness against multicollinearity than linear approaches. To interpret model behavior, SHapley Additive exPlanations (SHAP) analysis was applied, bridging the gap between data-driven predictions and plasma physics. Our analysis reveals that radio frequency power is the most influential parameter, while the low importance of gas flow rate indicates a surface-reaction-rate limited regime. Furthermore, DIC-derived features act as a high-resolution physical proxy for thin-film interference, providing complementary information to process parameters. Beyond accuracy, this framework establishes a physically meaningful correspondence between model-derived importance and underlying etching mechanisms. This approach provides a foundation for non-invasive, interpretable, and real-time process monitoring, enhancing transparency and control in semiconductor fabrication. To facilitate reproducibility, the experimental dataset is made publicly available.

I. INTRODUCTION

Precise control of plasma etching depth is essential for ensuring the performance and reliability of semiconductor devices. Traditional physics-based models, while capable of capturing fundamental plasma-surface interactions, often require complex, high-fidelity simulations that are computationally expensive and difficult to scale for real-time process monitoring¹. As a result, the practical application of such physics-based models in manufacturing remains limited, motivating the exploration of data-driven approaches for faster and more flexible process prediction. Building on this trend, recent studies have increasingly adopted data-driven modeling techniques to predict plasma etching behavior and optimize process conditions. Various machine learning (ML) techniques have been applied to accelerate design exploration and process optimization of plasma process. For instance, Guo et al.² trained a neural network on three-dimensional etching simulation data to optimize shallow trench isolation processes, while Ko et al.³ developed a deep learning-based framework

coupled with a particle swarm optimization algorithm for efficient plasma condition search. Kim et al.⁴ employed a Gaussian mixture model with optical emission spectroscopy data to accurately detect process endpoints, and Wan et al.⁵ as well as Lang et al.⁶ demonstrated Gaussian process-based models for dynamic sampling and sputtering optimization, respectively. Furthermore, Kim et al.⁷ implemented a deep learning-based reduced-order modeling framework to efficiently predict complex ion-surface interaction features, demonstrating the potential of data-driven models to bridge the gap between atomistic-scale physics and macroscopic process characteristics. More recently, Jung et al.⁸ introduced a heteroscedastic Gaussian process-based framework to quantify spatial variability and process-related uncertainties in SiO₂ etching, further enhancing the reliability of data-driven design optimization in semiconductor manufacturing.

In a similar context, our previous study⁹ demonstrated the feasibility of predicting etch depth solely from optical image data, establishing the potential of digital image colorimetry (DIC). The main motivation for incorporating optical data lies in its practical advantages—they can be acquired in real time and at low cost using non-contact imaging devices, which is critical for enabling continuous, cost-effective process monitoring and diagnosis in high-volume semiconductor manufac-

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turing environments. At the same time, because the optical-only framework does not explicitly incorporate controllable process parameters, it offers limited insight into how specific process conditions causally govern the observed etching behavior. Moreover, these studies rely heavily on black-box ML models that provide limited interpretability regarding the underlying physical mechanisms. This lack of physical insight hinders the direct application of models for fault diagnosis and proactive process control, which are critical for yield management. Furthermore, the high cost and time required to generate reliable experimental data pose a significant constraint. This necessitates a framework that can not only handle limited, expensive datasets but also effectively account for inherent uncertainty arising from machine variations and measurement errors.

Prior research has shown that studies in this field primarily rely on either process parameters or optical data in isolation. However, these two data sources are highly heterogeneous yet complementary: process parameters (e.g., power, pressure, flow) directly govern plasma dynamics (the *cause*), while DIC features reflect the post-etch surface condition (the *effect*) resulting from those processes. Nevertheless, their full potential remains unexplored, as the synergistic relationship between process parameters and optical changes has hardly been analyzed to the authors' knowledge. A key difficulty here is the methodological fusion of cause-related data (static process settings) and effect-related data (non-linear optical responses) to ensure the model learns their synergistic relationship.

To address these limitations, this study proposes an explainable machine learning framework that seamlessly integrates heterogeneous process and optical data to predict SiO₂ etch depth during plasma process. The framework is schematically described in Fig. 1.

The proposed approach is novel in four key aspects.

1. It combines two heterogeneous yet complementary data domains—controllable process parameters and DIC features—within a single predictive framework, enabling more accurate and robust estimation of etch depth.
2. To address the challenge of limited, high-cost experimental data and the inherent stochasticity in plasma processes (stemming from machine variability and measurement errors), a probabilistic treatment is introduced. Our approach will incorporate spatial and parametric variations along with simulated measurement uncertainties, allowing the predictive models to maintain a deterministic structure while effectively reflecting real-world input and output variability.
3. Multiple machine learning algorithms, including linear, neural network, and ensemble-based models are systematically compared to evaluate predictive performance and generalization capability under identical conditions.
4. Finally, explainable AI (XAI) techniques are employed to interpret feature contributions and reveal physically meaningful relationships between process and optical

variables, clearly distinguishing this study from conventional black-box modeling approaches.

Crucially, our explainable framework provides a quantitative, mechanism-based understanding of the complex coupling effects between process parameters, offering direct physical insights that are directly actionable for process engineers and control systems. Overall, this study extends data-driven plasma etch modeling into a new, multi-modal and interpretable domain, demonstrating the potential of combining quantitative physical interpretation of plasma dynamics with accessible optical information for transparent and practical AI applications in semiconductor manufacturing.

II. BACKGROUND

A. Plasma assisted etching

Plasma ion etching is a process that selectively removes material from solid surfaces in a directional manner by utilizing ions and reactive radicals produced within a plasma—a highly energetic, electrically charged gas phase¹⁰. This advanced method is extensively employed in the manufacturing of semiconductor components, display technologies, and high-aspect-ratio MEMS structures^{11–13}. For comprehensive background information, readers are directed to Ref.¹⁴, while recent progress in the field is summarized in Ref.¹⁵. Here, we provide only a concise overview of the fundamental physico-chemical mechanisms underlying plasma etching.

The plasma etching process starts by introducing a gas mixture—consisting of tetrafluoromethane (CF₄) and argon (Ar) for this study—into a vacuum chamber, where a strong radio-frequency (RF) electric field is applied. The applied RF energy ionizes the gas by dislodging electrons from the atoms and molecules, producing a luminous plasma made up of ions and free electrons. Within this plasma environment, CF₄ molecules decompose into reactive species such as fluorine atoms, while argon atoms become ionized but remain chemically inert. Material removal from the solid surface occurs through a combination of physical and chemical mechanisms.

Physically, positively charged ions in the plasma are accelerated toward the substrate under an electric bias, impacting the surface at high velocities and thereby assisting in material removal. This process resembles microscopic-scale sand-blasting, where each incoming ion collision dislodges minute fragments of the surface and disrupts its molecular bonds. Simultaneously, reactive fluorine atoms generated from the CF₄ plasma interact chemically with the substrate atoms—such as silicon in SiO₂—to produce volatile compounds like silicon tetrafluoride gas, which subsequently desorb from the surface. In summary, argon ions supply the energetic bombardment required for physical sputtering, while CF₄ contributes the chemically active fluorine species; the combination of these two effects results in a significantly faster and more efficient etching process compared to using argon alone. The general mechanism underlying the plasma-assisted etching process is schematically depicted in Fig. 2.

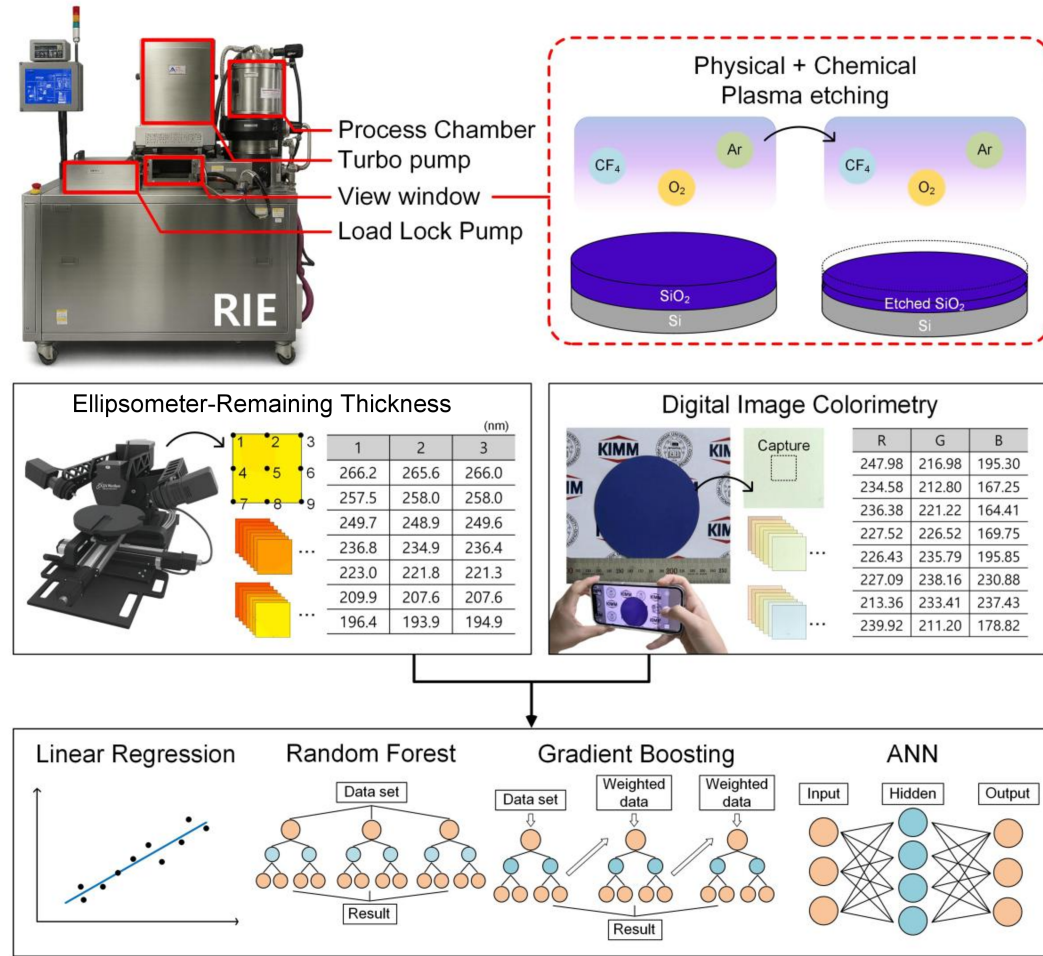


FIG. 1: Overall framework of the proposed modeling and analysis process.

The external process parameters directly influence the internal physical environment of the plasma. For instance, the RF power is the primary factor determining the plasma density; higher power increases the rate of ionization and dissociation, thereby providing more reactive radicals and ions for the etching process. The CF_4 flow rate acts as the primary source of these reactive species, determining the total amount of available fluorine radicals within the chamber. Meanwhile, the chamber pressure dictates the collision frequency between these species by defining the mean free path—the average distance a particle travels before colliding—which affects the energy transport to the substrate surface. These fundamental parameters work together to define the degree of material removal. Unfortunately, however, developing quantitative models that relate these parameters to macroscopic outcomes remains difficult due to system complexity and multiscale interactions. This limitation motivates the use of data-driven methods for improved process prediction.

TABLE I: Process conditions of ICP-RIE for SiO_2 etching

Parameter	Unit	Conditions
Chamber pressure	mTorr	20, 30, 40
Plasma power (Top)	W (watt)	50, 60, 70, 80, 90, 100, 110
Plasma power (Bottom)	W (watt)	20
Process time	sec	180
Gas (CF_4)	sccm	5, 10, 15, 20
Gas (Ar)	sccm	10
Gas (O_2)	sccm	10

B. Experimental protocols and measurement

In this study, we utilize the same experimental dataset as reported in our previous work⁹, which investigated plasma etch-

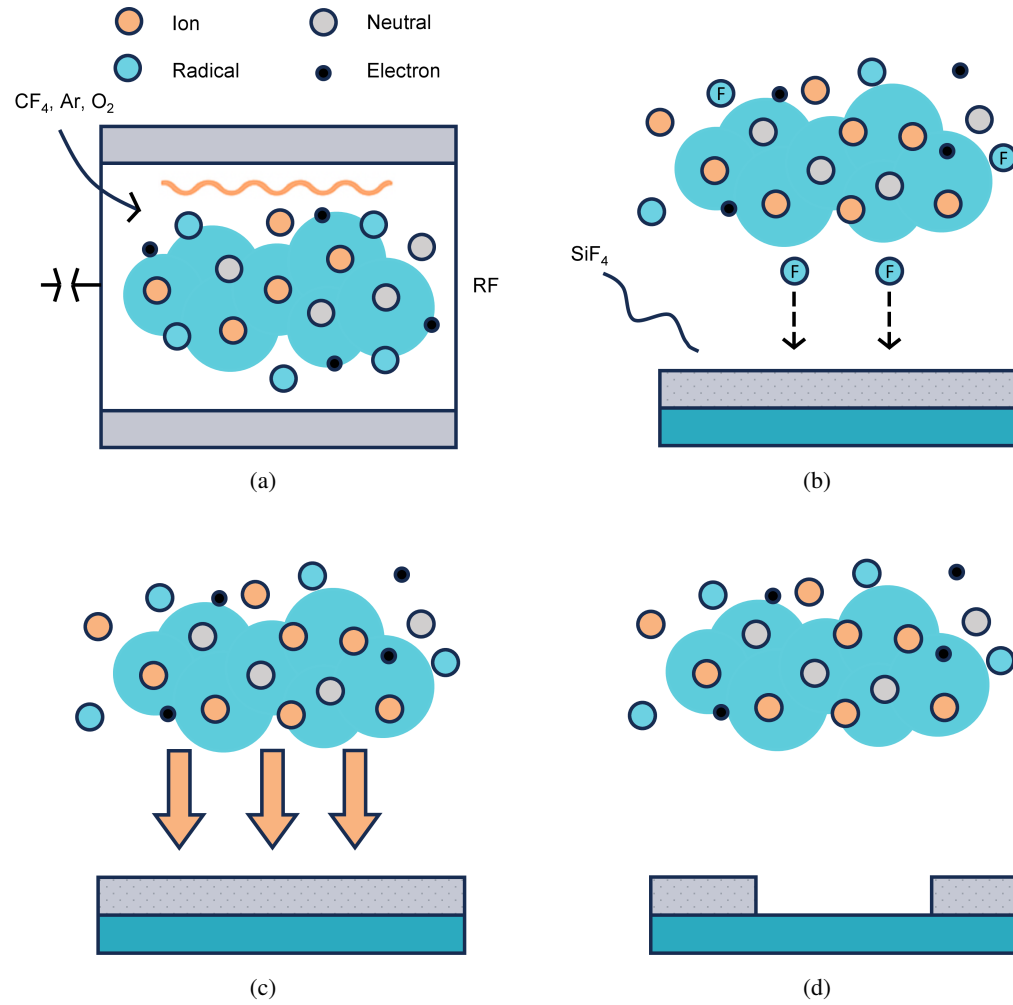


FIG. 2: Schematic illustration of the plasma etching process: (a) Plasma generation (b) Ion bombardment (c) Chemical reaction (d) Etching

ing processes under systematically varied operational conditions and used ANN and its Bayesian extension for providing machine-learning model predictions with uncertainties. The experiments for obtain the raw dataset were conducted using an inductively coupled plasma type reactive ion etcher (RIE; RAINBOW4420, Lam Research, Fremont, CA, USA). In this system, a high-density plasma is sustained by applying a 13.56 MHz radio frequency (RF) to the top copper coil. Optical emission spectroscopy (OES) was employed to examine the characteristic light emitted by the plasma, allowing identification of the atomic and molecular species produced during the etching process.

The plasma discharge was sustained using a gas mixture consisting of tetrafluoromethane (CF_4), argon (Ar), and oxygen (O_2). To maintain a baseline plasma environment, the gas flow rates of Ar and O_2 were kept constant at 10 sccm each and the bottom plasma radio frequency power is fixed at 20 W. On the other hand, the CF_4 gas flow rate ($\dot{Q}_{\text{CF}_4}$), chamber pressure (p), and RF power (P_{RF}) were selected as

the independent control variables and systematically varied to investigate their effects on the etching process. The uncertainties associated with them were estimated to be approximately 0.15 %, 1.0 %, and 0.5 %, respectively, based on the specification data provided by the equipment manufacturers. The range of all plasma process input parameters is summarized in Table I.

Each etching run was performed for 180 seconds under a specified set of input conditions, and the resulting SiO_2 etch depth was measured using a spectroscopic ellipsometer (M-2000V, J.A. Woollam, Lincoln, NE, USA). All samples had an initial SiO_2 thickness of 303 nm, and the etch depth was determined by subtracting the remaining film thickness after etching from the initial thickness. Thickness measurements were performed over an energy range of 1.0–3.5 eV, corresponding to wavelengths between 354 nm and 1240 nm. A total of nine repeated measurements were taken—one at the wafer center and eight located 0.5 cm from the center along the axial and diagonal directions—forming a nine-dimensional output vec-

tor that captures local etch variation.

After completing the ellipsometric measurements, additional color analysis was performed using images captured with a smartphone camera (iPhone 12, Apple Inc., Cupertino, CA, USA). To ensure maximum consistency in the color data acquisition—a critical step since color values are highly sensitive to illumination—strict protocols were implemented to control the photographic environment. Image acquisition was consistently performed in a controlled environment (e.g., a laboratory setting with minimal external daylight/window light) at a fixed time of day (to mitigate diurnal lighting shifts). The primary illumination was provided by a standardized white LED lamp. The white background served not only as a stable base but also as a diffuse reflector to ensure uniform light distribution across the sample surface, minimizing shadows. The distance between the camera lens and the sample surface, as well as the viewing angle, were mechanically fixed throughout the experiment to ensure consistent image scale and to guarantee a perpendicular viewing angle over the region of interest. Color data were extracted from a representative rectangular region of each sample, and the red (R), green (G), and blue (B) components ($0 \leq R, G, B \leq 255$) were quantified using ImageJ software. Fig. 3 presents the digital surface images of the total 84 plasma-etched samples and the corresponding SiO₂ thickness contours, indicating a clear correspondence between the optical appearance and the etched depth. Table II shows an example subset of the data, while the full dataset is provided in Appendix VI.

As shown in the data, the etch depth can be empirically controlled by adjusting process parameters. However, developing physical models that accurately relate these parameters to macroscopic outcomes remains challenging due to the multiscale and complex nature of the system. This limitation motivates the adoption of data-driven approaches to improve predictive modeling of the etching process. In particular, we aim to predict the etch depth using both process parameters and non-contact optical features (RGB data). These two sets of inputs are inherently different: process parameters directly govern the plasma physics, whereas RGB data represent indirect observations resulting from the process itself. Although RGB features are indirect, they are cost-effective, non-invasive, and available in real time, making them valuable auxiliary inputs for predictive modeling of the etching process.

C. Dataset characteristics and interpretation

The dataset utilized for training and validating the predictive models integrates two distinct yet complementary types of input features: controllable process parameters and optical response features derived from DIC. The successful integration and interpretation of these heterogeneous data sources are central to this study's methodological novelty.

The process parameters (i.e., $\dot{Q}_{CF_4}$, p , P_{RF}) are static, cause-related variables that are experimentally set to govern the plasma dynamics and etch rate. Conversely, DIC features (R, G, B) are effect-related variables that serve as a surrogate measure of the resulting post-etch surface condition. The do-

main of the latter parameters is inherently non-linear, primarily governed by the thin-film interference effect on the SiO₂ layer. The thin-film effect occurs when light interacts with silicon dioxide film—light waves reflecting off the top surface and the buried substrate interface interfere constructively or destructively, resulting in the observed color change. Crucially, the intensity of the reflected light (and thus the R, G, B values) is a non-linear, periodic function of the film thickness and the wavelength of the light. Moreover, plasma etching involves film thickness changes at the nanometer scale. Because such thin-film interference is exceptionally sensitive to these variations, the DIC features serve as an effective surrogate measure for predicting the final etch depth. While the inherent nanometer-scale sensitivity of the thin-film interference effect makes the DIC measures highly susceptible to minor experimental variations (e.g., ambient lighting or camera geometry), the reliability of our data is strongly affirmed by the results shown in Fig. 3. This figure demonstrates that the quantified optical data consistently follows the expected non-linear, periodic trends dictated by the physical mechanism of thin-film interference as a function of the known etch depth. The successful capture of this consistent relationship demonstrates that the strict experimental protocols employed (as detailed in Section II B) effectively mitigated experimental noise, supporting the use of DIC features as a reliable surrogate for etch depth monitoring.

Beyond the challenges posed by the data's heterogeneity and non-linear nature, we further recognize that the resulting dataset inherently contains physical multicollinearity. This coupling arises because process parameters do not influence the etch depth independently; rather, they collectively determine the intermediate plasma state variables, such as ion flux and ion energy, that drive the etching process. Moreover, the integration of DIC parameters introduces an additional layer of correlation, as they are highly correlated proxy for the output etch depth. This complex entanglement of cause and effect features in the input space is a major challenge for interpretation, as conventional black-box models may arbitrarily distribute predictive weight between these coupled parameters. To overcome this, we will adopt an Explainable AI technique in Section III B 2 to quantitatively decouple the feature contributions, allowing us to assess the pure, direct contribution of the controllable process settings on the final prediction.

III. METHODS

This section briefly outlines the models used in this study, their training procedures, validation methods, and performance evaluation criteria.

A. Regression models

A regression model is a statistical or machine learning model that predicts a continuous target variable based on given input features. All models ultimately solve a minimiza-

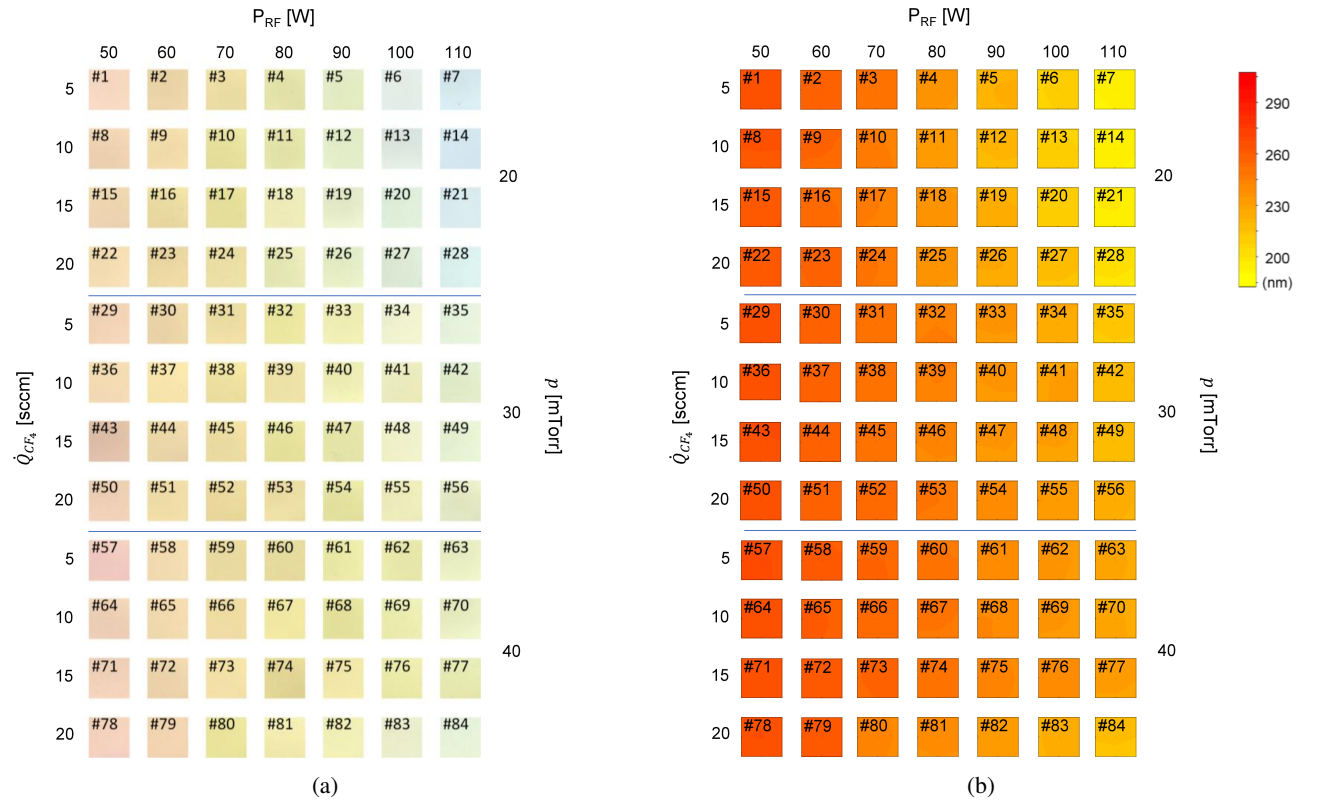


FIG. 3: Plasma etch samples: (a) digital surface images of SiO₂ coupon wafers (1.5 × 1.5 cm) and (b) corresponding SiO₂ thickness contours after etching, obtained from ellipsometry mapping at nine measurement points.

TABLE II: Example dataset comprising plasma etching process parameters, the corresponding RGB optical image and the corresponding remaining thickness of SiO₂.

Process Parameters			RGB			Remaining Thickness [nm]								
p [mTorr]	$\dot{Q}_{CF_4}$ [sccm]	P_{RF} [W]	R	G	B	T#1	T#2	T#3	T#4	T#5	T#6	T#7	T#8	T#9
20	5	50	248.0	217.0	195.3	266.2	265.6	266.0	266.2	266.6	266.7	266.4	266.0	265.8
20	10	90	227.0	237.1	200.5	221.4	220.7	221.1	220.0	219.8	216.9	219.0	218.9	220.8
30	10	60	248.4	230.4	173.4	256.8	256.1	255.9	255.2	256.0	256.1	256.9	257.1	256.9
30	15	60	237.2	216.4	169.8	259.4	259.7	259.9	259.6	259.0	258.2	258.0	257.7	259.1
30	20	60	239.7	221.7	168.1	258.4	257.9	258.9	258.7	258.5	257.8	257.5	256.7	257.5

tion problem of loss function $\mathcal{L}$ which is written as

$$\min_{\theta} \mathcal{L}(y, \hat{y}(\mathbf{x}; \theta)), \quad (1)$$

where θ denotes the model parameters (e.g., regression coefficients, neural network weights, or tree-splitting rules), $\hat{y}(\mathbf{x}; \theta)$ is the model prediction for a given input $\mathbf{x} = (p, \dot{Q}_{CF_4}, P_{RF}, R, G, B) \in \mathbb{R}^6$ ($n = 6$) represents the input features of the i -th sample. We consider models that predict the remaining SiO₂ thickness after 3 minutes of plasma etching. For the loss function $\mathcal{L}$, we consider the mean squared error (MSE) between the model predictions and the observed data

for N data samples, which are written as

$$\mathcal{L}(y, \hat{y}(\mathbf{x}; \theta)) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2. \quad (2)$$

1. Linear regression model

Linear regression (LR) is among the simplest and interpretable models for regression analysis. It is considered as

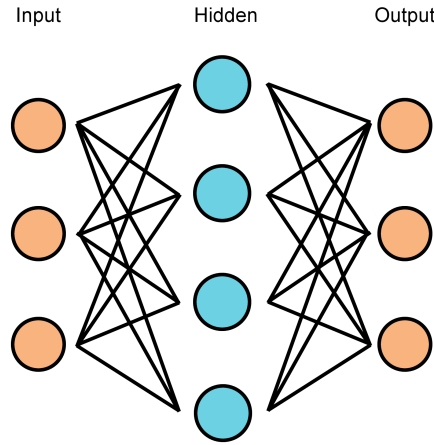


FIG. 4: Basic structure of an ANN

a baseline model. The general form of a multiple LR model

$$\hat{y}^L = \theta_0^L + \sum_{i=1}^n \theta_i^L x_i, \quad (3)$$

where $\hat{y}^L$ denotes the predicted target from LR model, θ_0^L is the intercept term, and θ_i^L represents the slope coefficient associated with the i -th feature x_i . When the MSE is adopted as the loss function, the optimal coefficients can be obtained analytically as:

$$\theta^L = (X^T X)^{-1} X^T y, \quad (4)$$

which is the exact solution that minimizes the MSE. In the above form, $X = [\mathbf{x}_1^T, \mathbf{x}_2^T, \dots, \mathbf{x}_N^T]^T \in \mathbb{R}^{N \times n}$ denotes the feature matrix.

Clearly, LR assumes a linear relationship between the input features and the target variable. While this assumption makes the model easy to understand and interpret, it also restricts its capacity to represent complex nonlinear interactions—such as those often observed in plasma etching processes. Nevertheless, LR remains a widely adopted baseline model owing to its simplicity and straightforward implementation.

2. Artificial neural network

Artificial neural network (ANN) is a computational model capable of capturing complex nonlinear relationships through hierarchical feature representations¹⁶. The architecture, illustrated in Fig. 4, consists of multiple layers where each connection is governed by weights and biases, transformed via non-linear activation functions¹⁷. In this study, we implemented a multi-layer perceptron (MLP) mapping input features $\mathbf{x}$ to the target output through a series of linear transformations and activation layers. The training process involves optimizing the model parameters—weights and biases—to minimize a loss function $\mathcal{L}$ (4) using backpropagation and gradient-based optimizers^{18,19}. To ensure optimal performance and

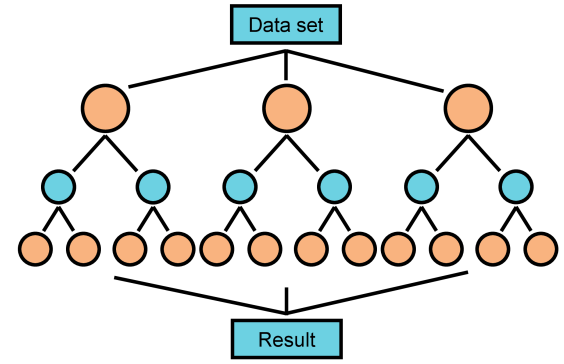


FIG. 5: RFM model structure and data processing flow

prevent overfitting, key hyperparameters, including the number of hidden layers, neurons per layer, and the learning rate, should be carefully tuned.

3. Random forest model

Random forest model (RFM) is an ensemble learning method that combines predictions from multiple decorrelated decision trees to improve robustness and reduce variance²⁰. The model architecture, schematically shown in Fig. 5, involves recursively partitioning the feature space through binary splits. The final prediction $\hat{y}$ is obtained by averaging the outputs of T individual trees, which effectively mitigates the overfitting common in single decision trees. The training process employs bootstrap aggregating (bagging) and random feature selection to ensure diversity among the trees. Specifically, each tree is grown independently using a random subset of the training data and a random selection of input features at each node split. This approach allows the RFM to remain robust against high-dimensional data and noise, while alleviating the 'curse of dimensionality'²¹. Key hyperparameters, such as the number of trees and the maximum tree depth, should be optimized to achieve the best generalization performance.

4. Gradient boosting

Gradient boosting (GB) is an ensemble technique that builds a strong predictive model by sequentially adding weak learners, typically decision trees, to correct the errors of previous iterations²². The model architecture is illustrated in Fig. 6. Unlike the independent training in RFM, GB follows an additive process where each new learner $w_m(\mathbf{x})$ is fit to the negative gradient of the loss function, controlled by a learning rate. This gradient-based optimization allows GB to progressively reduce the loss through iterative refinement. A key advantage of GB is its flexibility in optimizing various loss definitions and regularization strategies, such as subsampling and shrinkage, to improve generalization²³. Compared to ANN,

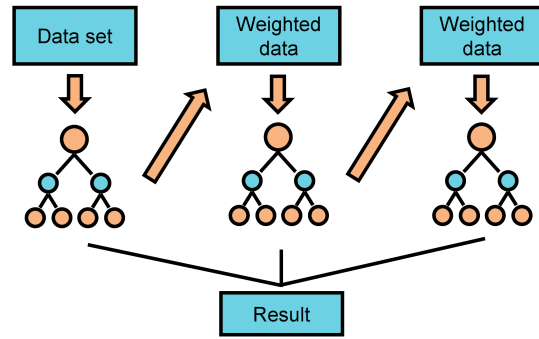


FIG. 6: GB model structure and data processing flow

GB allows for easier customization of loss functions without architectural changes, and it offers finer control over the optimization process than RFM. While this flexibility often leads to superior predictive performance on well-tuned datasets, it requires more careful hyperparameter tuning and can be sensitive to noise²⁴

B. Model validation and analysis

Here, we describe the validation framework used to evaluate and compare the performance of different models. Common evaluation metrics and interpretability methods (e.g., SHAP) are employed to ensure consistent and reliable model assessment.

1. K-fold cross-validation and evaluation metrics

To evaluate the reliability and generalization performance of the model, K-fold cross-validation was employed. In this technique, the dataset is divided into K equal subsets; each subset is used once for validation while the remaining $K - 1$ subsets are used for training. Repeating this process K times allows every sample to be validated exactly once, providing an unbiased estimate of model performance.

The predictive accuracy of the models was assessed using three commonly used evaluation metrics:

1. **Root Mean Square Error (RMSE)** measures the square root of the average squared difference between predicted and actual values.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

2. **Mean Absolute Error (MAE)** represents the average absolute deviation between predicted and actual values.

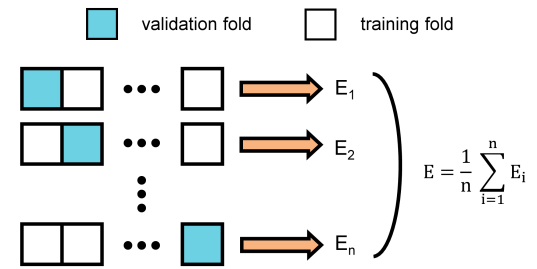


FIG. 7: A schematic illustrating K-fold cross-validation process

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n \quad (6)$$

3. **Coefficient of Determination (R^2)** indicates how well the model explains the variance in the observed data; values close to 1 imply better predictive performance.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad \text{where} \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad (7)$$

For RMSE and MAE, smaller values indicate better model performance as they represent lower prediction errors. In contrast, an R^2 value closer to 1 implies that the model explains a larger proportion of the variance in the observed data.

The overall performance of each model was estimated using the average of the evaluation metrics across all folds. Since the dataset is relatively small, the K-fold cross-validation approach ensures a more stable and unbiased evaluation of model performance. This procedure is schematically illustrated in Fig. 7.

2. SHAP analysis

In this study, we go beyond mere prediction by employing analysis techniques to explore the underlying relationships among plasma processes, DIC parameters, and etch depth. This approach aims to provide deeper insights into these interactions and ultimately contribute to the broader development of XAI in this field. To this end, we employ SHAP analysis to help interpret and better understand the factors influencing the model predictions.

The foundation of this method lies in cooperative game theory^{25,26}, where the objective is to fairly distribute pay-offs among players according to their individual contributions to the overall outcome. In the context of machine learning, SHAP values are used to assign an importance score to each feature based on its contribution to the model's prediction, thereby providing a consistent framework for feature attribution^{25,27}.

To illustrate, given a model prediction $\hat{y}$ that takes into account a total n features, the SHAP value ϕ for i -th feature is defined as follows²⁵:

$$\phi_i = \sum_{S \subseteq M \setminus \{i\}} \frac{|S|!(n-|S|-1)!}{n!} [\hat{y}(S \cup \{i\}) - \hat{y}(S)] \quad (8)$$

where M denotes the set of all features, S is any subset of features excluding i , $|S|$ represents the number of features in S , and $\hat{y}(S)$ denotes the model's prediction using features in subset S . Note that the weighting factor in Eq. (8) ensures a fair attribution of the impact of each feature considering all possible feature coalitions, where a coalition refers to any subset of features that jointly contribute to the model's prediction.

SHAP provides both global and local interpretability of machine learning models. Global interpretability refers to understanding which features are most influential across the entire dataset, thereby offering insights into the general patterns the model has learned (e.g., ranking features by their average contribution to predictions). Local interpretability, in contrast, focuses on explaining individual predictions by quantifying how much each feature contributes to a specific output, allowing users to understand why the model produces a particular result. Because SHAP derives feature contributions from the way a model actually learns from data, feature importance may naturally differ across models depending on their assumptions and architectures. For instance, a feature that exhibits a strong linear relationship with the target variable may appear highly important in a linear regression model, but less so in a RFM if nonlinear interactions dominate the prediction.

IV. MODELING FRAMEWORK AND IMPLEMENTATION DETAILS

Next, we describe the overall modeling framework and implementation details used in this study, including the underlying data assumptions, uncertainty treatment, and training configurations.

A. Data and modeling assumptions

To begin with, we recall that the complete dataset, as described in Appendix VI, consists of 84 samples in terms of input variables. Because the experimental data are expensive to acquire, only a limited number of samples are available. Moreover, measurements collected under identical process conditions exhibit variability across different spatial locations, indicating inherent uncertainty in the data. To maximize the utility of the available samples, we explicitly incorporate the spatial variability inherent in the thickness measurement process. Specifically, measurements were collected from nine distinct spatial locations under each identical process condition, resulting in a total of $84 \times 9 = 756$ empirical data points. Rather than relying on averaged values, all individual measurements were retained to preserve the intrinsic output variability of the etching process.

Because the output measurements result in nine realizations of the response variable, a corresponding representation of the input variables was required to match the size of the dataset. To achieve this in a physically meaningful manner, we considered instrumental uncertainties in the input variables, thereby introducing input variability consistent with realistic measurement tolerances. This treatment allows the full set of measurement outcomes to be utilized while preserving the underlying physical consistency of the process. To this end, each of the 84 nominal process conditions was augmented by generating nine perturbed input instances using independently sampled Gaussian noise. Based on the equipment manufacturer's specifications and the accuracy limits of the optical imaging system, measurement tolerances were assigned to each input feature (e.g., $\pm 0.5\%$ for $\dot{Q}_{\text{CF}_4}$, $\pm 0.15\%$ for p , $\pm 0.01\%$ for P_{RF} from the nominal values and ± 0.1 for each RGB color), and the standard deviations of the Gaussian perturbations were chosen accordingly. This procedure yielded a total of $84 \times 9 = 756$ input-output data pairs, enabling the model to be trained on a dataset that consistently reflects both spatial output variability and realistic instrumental input uncertainty, while preserving the underlying deterministic process structure. Consequently, the extracted physical insights, including those obtained through SHAP analysis, are grounded in input-output relationships informed by the experimentally observed physical variability of the system.

We note that the probabilistic perturbation of input variables was introduced to enable a consistent and robust model evaluation framework under data-scarce experimental conditions, particularly for data-driven models such as ANN, and does not alter the qualitative physical conclusions of this work.

B. Implementation details

All machine learning algorithms were implemented using PyTorch²⁸, while data preparation steps utilized the Pandas library. The input features are defined as $\mathbf{x} = (p, \dot{Q}_{\text{CF}_4}, P_{\text{RF}}, R, G, B)$, where p is the chamber pressure, $\dot{Q}_{\text{CF}_4}$ is the CF_4 gas flow rate, P_{RF} is the radio-frequency power, and R , G , and B represent the color intensity values extracted from the DIC images. The target variable y corresponds to the remaining thickness of the SiO_2 layer after plasma etching.

Each component of $\mathbf{x}$ and y were utilized in their original physical scales during the training process. Since the input variables were within a comparable order of magnitude (ranging from 10^1 to 10^2 , see full dataset in Appendix VI.), additional feature scaling or standardization was not applied.

To ensure fair and reproducible comparisons among different models, random seeds were fixed across all data post-processing procedures, including the addition of input uncertainties and data mixing. This setup allowed each sample to be consistently used for both training and validation, enabling stable estimation of the overall predictive performances of different models.

1. ANN

ANN with dropout was implemented to model the nonlinear relationship between input characteristics and etch depth. The network consisted of two hidden layers with the number of neurons varied among (8–4), (16–8), (32–16), and (64–32) configurations during hyperparameter exploration, resulting in a total of four fully connected layers including the input and output layers. A rectified linear unit (ReLU) activation function was applied to the hidden layers, and a dropout probability of 0.1 was used to mitigate overfitting. The model was trained using mini-batch gradient descent (batch size of 32) for up to 2000 epochs. Both Adam and AdamW optimizers were evaluated, with learning rates of 0.1, 0.01, and 0.001, and weight decay values of 0.1, 0.01, and $1e-4$. AdamW demonstrated more stable convergence and was adopted in the final configuration, with a learning rate of 0.001 and weight decay of $1e-4$. Early stopping with a patience of 100 epochs was applied to prevent overfitting, and the best model parameters were restored during training.

2. RFM

Hyperparameter optimization of the RFM was performed using random search. The number of trees was tested at 100, 200, and 300 to balance accuracy and computational cost. The maximum tree depth was explored at 10, 20, and 30, as well as an unrestricted setting. The minimum number of samples required for node splitting was set to 2 or 5, and that for leaf nodes was set to 1 or 2 to smooth predictions and mitigate overfitting. The number of features considered at each split was set to $\sqrt{\text{features}}$ and $\log_2$, aiming to reduce correlation between trees and enhance generalization.

3. GB

The optimization of the hyperparameters of the GB model was also performed by random search. The number of boosting steps was set to 100, 200, and 300. The learning rate was set to 0.01, 0.05, and 0.1 to adjust the contribution of each tree. The tree maximum depth was set in the range of 3–7 to balance pattern learning ability and overfitting prevention. The minimum number of samples for internal node division was set to 2 or 5, and the minimum number of samples for leaf nodes was set to 1 or 2, softening the prediction and enhancing model stability. This hyperparameter range setting was selected to maintain generalization performance while learning meaningful trends in the data.

V. RESULT AND DISCUSSION

A. Quantitative performance evaluation

The prediction results for each fold are visualized in Fig. 8. In this figure, the predictions of each model across the five

folds are plotted together. The vertical axis represents the data index, while the horizontal axis denotes the remaining thickness of the plasma etching process. The black line indicates the actual measured values, whereas the model predictions are shown as scatter points. For clearer comparison, the data within each fold are sorted by etch depth. The horizontal deviation of the scatter points from the black line corresponds to the prediction error. Overall, all machine learning models capture the general trends more accurately than the linear models.

The quantitative evaluation indicators of RMSE, MAE, and R^2 are summarized in Table III. First, the LR model, despite exhibiting relatively stable performance RMSE: 2.697 ± 0.182 , and R^2 : 0.979 ± 0.003 , showed clearly lower accuracy than machine learning models. This indicates that the underlying relationship between optical properties and etching depth is inherently nonlinear. The ANN demonstrated better predictive accuracy, having an R^2 value of 0.991 ± 0.001 , but slightly lower than the ensemble models (RFM and GB). However, its prediction error was slightly higher than those of the ensemble models, which can be attributed to the intrinsic sensitivity of neural networks to hyperparameter settings and data preprocessing²⁹.

The ensemble models exhibited stable generalization performance, producing predictions that closely matched the actual values. In particular, the RFM model achieved the lowest prediction error with an RMSE of 1.369 ± 0.110 and an MAE of 1.072 ± 0.061 , indicating consistent performance across folds. The GB model yielded a comparable level of accuracy, further confirming the robustness of ensemble learning methods over linear or neural network approaches. However, since no significant performance improvement of the GB model over the RFM was observed across various parameter settings, the subsequent analysis considers the RFM as the representative ensemble method.

B. Feature importance and multicollinearity analysis

This section presents the results of the SHAP analysis, focusing on the quantitative assessment of feature importance across different models. Furthermore, we leverage these results to analyze the impact of physical multicollinearity inherent in the integrated dataset.

First, Fig. 9 shows the SHAP summary scatter plots, which are also known as beeswarm plots, for different models—LR,

TABLE III: Comparative performance of regression models

Model	RMSE [nm]	MAE [nm]	R^2	TOP
Linear	2.697 ± 0.182	2.154 ± 0.151	0.979 ± 0.003	G
ANN	1.978 ± 0.171	1.524 ± 0.158	0.989 ± 0.002	P_{RF}
RFM	1.369 ± 0.110	1.072 ± 0.061	0.995 ± 0.001	P_{RF}
GB	1.380 ± 0.111	1.080 ± 0.063	0.995 ± 0.001	P_{RF}

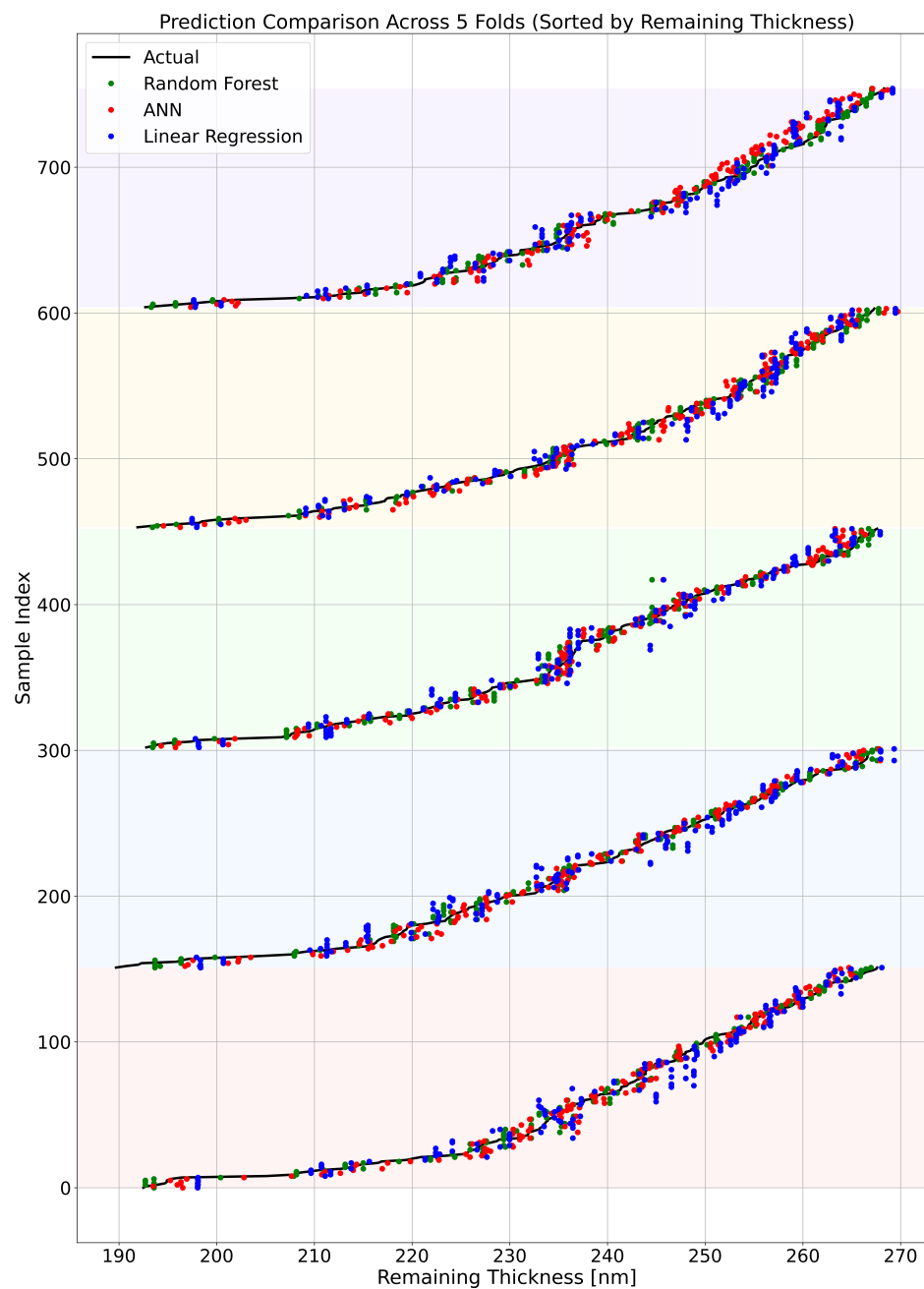


FIG. 8: Model prediction comparison across 5 folds. Different folds have different region specific colors.

ANN, and RFM. The GB model was excluded from this comparison, as it did not show any notable improvement over the RFM. In addition to the visual interpretation, Table IV summarizes the quantitative SHAP analysis results for each regression model, listing the average absolute SHAP values and corresponding feature-importance rankings.

In Fig. 9, each dot corresponds to a single data instance, representing the local SHAP value for that feature. The horizontal axis denotes the SHAP value, which quantifies the impact of that feature on the predicted output. A positive SHAP value indicates that the feature increases the predicted

etch depth, whereas a negative value means the feature contributes to lowering it. The color of each point encodes the feature's original value (red for high and blue for low), the scale of which is determined within each feature based on its range in the data set. Thus, the colored points reveal how the magnitude of a feature influences the direction of its effect on the prediction. The vertical axis lists the individual features, which are arranged according to their importance ranking for each model, allowing direct comparison of their relative contributions. The horizontal spread of points for each feature reflects its overall importance to the model. Features

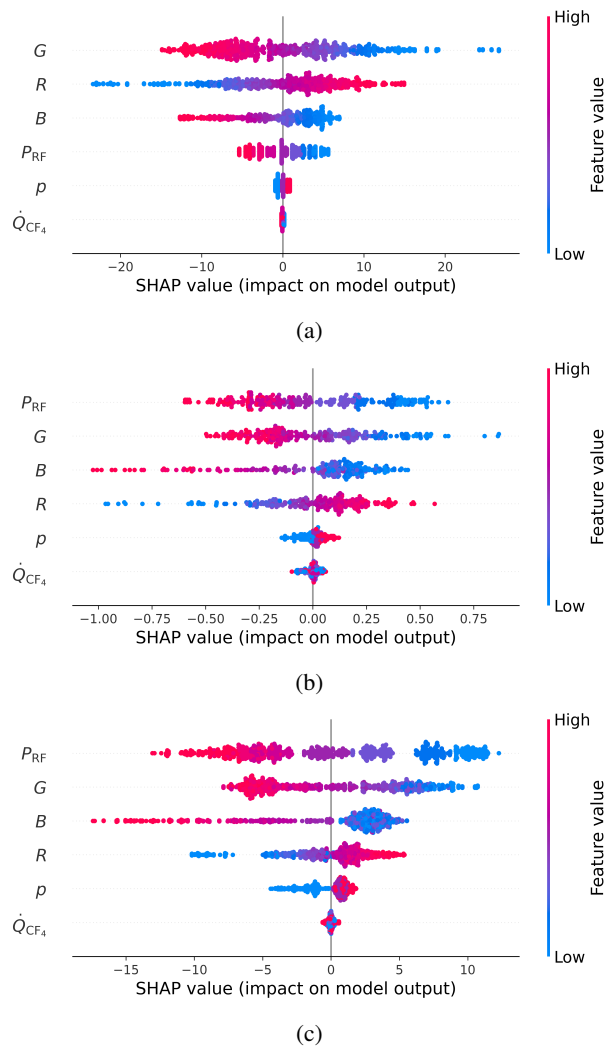


FIG. 9: The SHAP summary scatter plots (beeswarm plots) for (a) Linear model, (b) ANN and (c) RFM.

whose SHAP values are widely distributed across both positive and negative directions exert a strong influence on model predictions, whereas those with values concentrated around zero have little effect. Since SHAP calculates feature contributions based on how a model has actually learned from the data, it is natural that feature importance may differ across models depending on their structures. For example, this is evident in the substantially different feature-importance rankings observed between the linear model and the other models (RFM and ANN).

It is important to note that when comparing SHAP results across different model architectures, the absolute magnitudes of the SHAP values are not directly comparable, as they are highly dependent on the internal mathematical formulation of each model²⁵. As shown in Fig. 9, for instance, the SHAP values for the ANN are distributed within a significantly narrower range (approximately -1.0 to $+0.8$) than those of the Linear and Random Forest models. This discrepancy is primarily attributed to the intrinsic architecture of the ANN; specifi-

cally, the internal weight regularization and the nonlinear activation functions tend to numerically compress the contribution of each feature during the backpropagation process³⁰. Therefore, the focus of cross-model comparison should remain on the relative importance rankings of the features rather than their absolute numerical values. The fact that the importance rankings remain consistent across all models—despite these variations in scale—confirms that the ANN, ensemble, and linear approaches all successfully and robustly capture the same underlying physical dependencies of the plasma process.

Several common characteristics are observed across all models in Fig. 9 and Table IV. First, pressure p and the flow rate of CF_4 ($\dot{Q}_{CF_4}$) are classified as the two most unimportant features. Also, among optical features, the green color is considered as one of the most important features for correctly predicting the etch depth of the plasma process. On the other hand, there is a clear difference between the results of the linear model and those of the other models. In the linear model, the importance of plasma power P_{RF} is significantly underestimated compared to other models. This discrepancy arises primarily from multicollinearity among the input features. To illustrate, while the process parameters— p , $\dot{Q}_{CF_4}$, P_{RF} —are independently controlled experimental variables, the optical features (R, G, and B) extracted from post-etching images are outcomes influenced by these process conditions. Thus, it is highly plausible that strong correlations exist between the process and optical variables, making a certain degree of multicollinearity among the input features unavoidable. In the presence of such multicollinearity, the coefficient estimates in Eq. (4) become unstable and unreliable due to the ill-conditioning of the feature matrix X ³¹. Consequently, although the overall prediction accuracy of the linear model may remain acceptable, the interpretability and generalization capability of the model can be severely compromised.

In contrast, the nonlinear ANN and RFM ensemble models are less sensitive to such correlations and are therefore capable of capturing more complex feature dependencies, resulting in a more robust assessment of feature importance^{32,33}. The feature-importance rankings obtained from the RFM and ANN models closely validate each other, supporting the reliability of these estimates. It is also noted that the feature-importance ranking in the ANN model varies slightly across different hyperparameter configurations, reflecting the model's sensitivity to initialization and training conditions^{34,35}. Despite such variations, both models consistently identify P_{RF} as the most influential process parameter, while the G–B–R values extracted from DIC data also contribute significantly to etch-depth prediction. This finding suggests that combining process parameters with DIC-derived features provides an effective strategy for etch-depth prediction, as the two sets of variables complement each other.

C. Physical Interpretation of Feature Contributions

The quantitative feature contribution derived from the SHAP analysis not only validates the predictive utility of the input parameters but also offers mechanism-based physical in-

TABLE IV: SHAP analysis results for regression models

Model	1	2	3	4	5	6
LR	G	R	B	P_{RF}	p	$\dot{Q}_{CF_4}$
	+ 6.72	+ 5.76	+ 3.75	+ 2.61	+ 0.49	+ 0.06
ANN	P_{RF}	G	B	R	p	$\dot{Q}_{CF_4}$
	+ 0.27	+ 0.24	+ 0.23	+ 0.2	+ 0.06	+ 0.02
RFM	P_{RF}	G	B	R	p	$\dot{Q}_{CF_4}$
	+ 5.87	+ 4.72	+ 4.09	+ 2.28	+ 1.17	+ 0.13

sights into the plasma etching process, moving beyond mere correlational observations.

Firstly, the high importance attributed to P_{RF} directly reflects its role as the primary energy source governing the plasma. An increase in P_{RF} enhances the electron density and temperature, leading to a greater rate of dissociation and ionization. This results in an increase in both the reactive radical density (critical for chemical etching) and the ion flux and energy (critical for physical etching/anisotropy). The SHAP values thus confirm that power is the most significant and direct factor for controlling the overall etch rate. Similarly, the substantial importance of the DIC features (R, G, and B) is physically justified by the nanometer-scale sensitivity of the thin-film interference effect. The R, G, B values act as highly reliable, real-time indicators of the optical path difference in the SiO_2 layer, confirming that the physical phenomenon governing the color change is another deterministic input for predicting the resulting etch depth. The integrated SHAP results thus provide a comprehensive, physics-informed decomposition of the etching system, validating the complex interplay of energy input, particle transport, and optical response.

Conversely, the consistently low importance assigned to the $\dot{Q}_{CF_4}$ and p provides crucial validation of the underlying plasma physics within the tested process window. The low contribution of $\dot{Q}_{CF_4}$ strongly suggests that the experiment operated primarily in a reaction-rate limited regime rather than a supply-limited regime³⁶. Since the etchant gas supply was sufficient, the etch rate was predominantly governed by the energy input (P_{RF}) and subsequent radical generation. Also, although p is known to influence the anisotropy by changing the mean free path of ions, its low SHAP value suggests that the change in etch rate across the tested pressure range was secondary compared to the effects of power and the optical response.

This comprehensive analysis, encompassing both the most and least influential factors, confirms that the machine learning model successfully distilled the underlying plasma physics from the experimental data.

VI. CONCLUSION

In this work, we developed a non-contact etch-depth prediction framework that integrates both plasma process parameters and optical features obtained from DIC imaging during plasma etching. Unlike optical-only models, which primarily capture phenomenological correlations, the integrated framework employed in this study provides insights into how process parameters influence etching behavior. By combining these two complementary data sources, the model captures both controllable process conditions and their resulting physical manifestations on the substrate surface. Among the evaluated machine learning algorithms, the ensemble-based RFM achieved the highest predictive accuracy, demonstrating its robustness in handling high-dimensional, multi-modal datasets.

The SHAP-based explainability analysis provided both statistical and physical validation of the model. We identified that while a linear model suffers from instability due to multicollinearity between process and optical variables, nonlinear models like RFM and ANN effectively capture complex feature dependencies. Specifically, P_{RF} was identified as the most dominant process parameter, physically corresponding to the energy-driven generation of ions and radicals. Furthermore, the low importance of $\dot{Q}_{CF_4}$ suggested that the process operated in a surface-reaction-rate limited regime, a finding that aligns with fundamental plasma physics.

The significant contribution of DIC-derived G–B–R features further confirms the utility of thin-film interference as a high-resolution proxy for etch-depth monitoring. These findings demonstrate that integrating optical imaging with AI-based modeling offers a non-invasive, physically interpretable, and real-time diagnostic tool for plasma processes.

While the present results (ensemble modeling and SHAP analysis) provides convincing physical insights into the plasma etching process, it is also important to note that these interpretations are primarily valid within the explored parameter range and the specific substrate materials used. When generalizing our modeling approach to other process windows or different materials, one must consider the non-linear shifts in plasma chemistry and surface interactions. For instance, a change in substrate material may alter the dominant reaction pathways, potentially shifting the significance of specific optical parameters identified in this study. Future work will focus on cross-material validation to establish a more universal, explainable framework for plasma etch processing.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Jihoon Myung: Formal analysis, Investigation, Writing – original draft. **Eunseo Go:** Investigation, Methodology **Donghyun Cho:** Investigation, Methodology. **Minkyung Ko:** Formal analysis. **Nari Jeon:** Resources. **Jaehyun Kim:** Validation. **Bongsu Hahn:** Formal Analysis, Validation. **YongSu Jung:** Validation, Methodology **Hyeong-U Kim:** Conceptualization, Supervision, Project administration, Formal analysis. **Jaekwang Kim:** Supervision, Formal analysis, Writing – review & editing.

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DATA AVAILABILITY

The data that supports the findings of this study are available within the article.

AUTHOR DECLARATIONS

Conflict of interest

The authors have no conflicts to disclose.

Ethics approval

Ethics approval was not required for this study as it did not involve human participants or animals.

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FULL DATA

TABLE V: Raw data set representing etched SiO₂ thickness (Part I)

Process Parameters			RGB			Etched Thickness [nm]								
p	$\dot{Q}_{CF_4}$	P_{RF}	R	G	B	T#1	T#2	T#3	T#4	T#5	T#6	T#7	T#8	T#9
20	5	50	248	217	195.3	266.2	265.6	266	266.2	266.6	266.7	266.4	266	265.8
20	5	60	234.6	212.8	167.3	257.5	258	258	257.8	257.3	256.5	256.7	256.7	257.4
20	5	70	236.4	221.2	164.4	249.7	248.9	249.6	249.7	249.4	248.5	248.5	247.6	248.8
20	5	80	227.5	226.5	169.8	236.8	234.9	236.4	236.6	236.7	235.6	235.5	234.2	235.3
20	5	90	226.4	235.8	195.9	223	221.8	221.3	218.8	220.1	219.4	222.1	222.7	223.4
20	5	100	227.1	238.2	230.9	209.9	207.6	207.6	204.7	206.6	205.1	208.3	208.2	209.5
20	5	110	213.4	233.4	237.4	196.4	193.9	194.9	191.9	192.8	189.7	192.7	192	194.9
20	10	50	239.9	211.2	178.8	262.7	263.3	263.5	263.3	262.5	261.6	261.7	261.5	262.6
20	10	60	241.3	221	171.4	255.4	255.2	255.7	255.8	255.5	254.7	254.4	253.7	254.7
20	10	70	230.6	224.4	157	245.8	243.4	244.8	245.7	246.5	246.9	246.1	244.8	244.3
20	10	80	227.2	228.4	170.1	234.7	234.2	234.9	234.9	234.4	233.1	233.1	232.6	233.6
20	10	90	227	237.1	200.5	221.4	220.7	221.1	220	219.8	216.9	219	218.9	220.8
20	10	100	224.2	235.1	225.3	210.2	205.3	207.9	207.9	209.9	209.3	209.2	207	207.5
20	10	110	211.6	230.6	238.6	196.4	192.8	195.3	194.7	195	192.5	193.5	190.7	193.4
20	15	50	238.3	214.1	176.5	262.2	262.1	261.9	261.5	261.7	261.8	262.3	262.6	262.4
20	15	60	234.9	219.7	161.7	253.4	251.6	252.6	253.3	254.1	254.5	253.6	252.5	252.3
20	15	70	230.3	223.3	155.3	254.2	243.1	245.4	246.6	246.2	245.3	244.4	242.8	243.5
20	15	80	238.9	235.9	185.9	236.9	234.4	235.3	235.7	237.2	237.6	237.3	236.4	235.8
20	15	90	228.7	235.1	201.3	223.7	221.1	221.7	221.3	223	223.5	223.9	222.9	222.5
20	15	100	223.6	239.1	218.6	212.1	212	212.2	211	210.5	208.1	209.3	208.6	211.3
20	15	110	217.8	235.3	239	198.2	195.9	198	197.4	197.2	194	194.5	192.3	195.6
20	20	50	244.6	221.7	180	262.1	262.2	262.5	262.4	262.2	261.6	261.4	261	261.8
20	20	60	236.8	216.6	163.5	256.1	257.1	256.4	255.3	255.1	254.5	255.5	256.1	256.7
20	20	70	236.4	224.5	164.5	247.6	246.2	247.5	248	248.1	248	247.3	246.1	246.4
20	20	80	231.4	232.8	181.9	236.8	236.8	237.6	237.3	236.4	234.9	235	234.3	236
20	20	90	229.8	238	198.7	227.9	226.8	228.1	228.1	227.7	225.7	225.7	223.6	226.4
20	20	100	220.9	234.4	213.5	217.2	216	217.3	217.2	216.6	214.4	214.4	212.8	215.4
20	20	110	219.5	241.4	238.3	202	199.7	201.8	201.9	201.4	198.2	198.5	196.4	199.8
30	5	50	242.9	212.1	182	266.7	267.2	267.5	267.6	266.7	265.7	265.7	265.4	266.4
30	5	60	236.1	214.2	168	258.4	258.8	258.2	257.1	257.2	256.8	257.9	258.5	258.9
30	5	70	237.9	222.5	163.6	251.5	251.9	251.8	251.1	250.7	249.8	250.5	250.1	251.2
30	5	80	236.2	231.8	160.7	244.9	242.7	243	242.3	244.2	245	245.7	245.1	244.4
30	5	90	239.7	238.6	181.6	236.4	237	237.3	236.7	235.6	233.2	233.9	233.6	235.8
30	5	100	240.6	242.7	208.6	227.3	227.6	228.2	227.6	226.3	223.8	224.4	223.5	226.4
30	5	110	229.3	243.4	221.4	214.9	214.9	215.1	213.7	213.2	210.6	212.3	211.5	214.2
30	10	50	244.2	220.2	180.1	264.6	263.8	264.7	265.1	264.9	264.1	263.9	263	263.6
30	10	60	248.4	230.4	173.4	256.8	256.1	255.9	255.2	256	256.1	256.9	257.1	256.9
30	10	70	239	225.5	161.1	250.6	250.2	250.9	250.9	250.3	249.1	249.3	248.7	249.8
30	10	80	238.3	226.9	166.7	245.1	245.1	245.2	244.6	244.5	243.2	243.8	243.5	244.7
30	10	90	237.8	237.8	181	236	236.7	236.6	235.3	234.6	232.7	233.9	234	236
30	10	100	236.1	236.7	192.4	231.1	229.1	228.9	227	228.5	228.7	231.1	231.9	231.8
30	10	110	225	236.1	203.2	219.5	219.9	219.1	216.6	216.4	214.8	217.5	218.4	220.1
30	15	50	224.5	192.9	169.7	266.6	267.3	267.1	266.5	266.1	265.2	265.6	265.7	266.8
30	15	60	237.2	216.4	169.8	259.4	259.7	259.9	259.6	259	258.2	258	257.7	259.1
30	15	70	234.5	219.5	160.6	251.6	252	252	251.4	251	249.9	250.3	250.2	251.5

TABLE VI: Raw data set representing etched SiO₂ thickness (Part II)

Process Parameters			RGB			Etched Thickness [nm]								
p	$\dot{Q}_{CF_4}$	P_{RF}	R	G	B	T#1	T#2	T#3	T#4	T#5	T#6	T#7	T#8	T#9
30	15	80	230.6	226.1	156.6	243.9	244.2	243.9	243.1	242.6	241.3	242.3	241.9	243.9
30	15	90	229.2	229.5	168.4	235.7	235.2	236.1	235.8	235.3	233.5	233.7	234.1	234.9
30	15	100	240.5	241.2	201.9	230.8	229.6	231.4	233	230.3	228	228.2	226.9	229.3
30	15	110	233.5	244.5	213.7	219.8	219.5	220.2	219.4	218.8	216.9	217.3	216.2	218.6
30	20	50	237.1	207	178.8	265	265	264.7	263.9	264.5	264.1	264.3	264.2	264.9
30	20	60	239.7	221.7	168.1	258.4	257.9	258.9	258.7	258.5	257.8	257.5	256.7	257.5
30	20	70	236	220	159.3	254.2	252.8	254.1	254.6	254.6	253.9	253.4	252.1	252.9
30	20	80	236.3	221.3	162.4	247.7	245.9	247.4	248.1	248.2	247.3	246.7	245	245.9
30	20	90	228.1	227.1	161.5	240.6	241.2	241	239.9	239.3	237.3	238.5	238.8	240.6
30	20	100	238.3	236.3	186.5	234.5	231.1	233.5	234.5	235.2	234.6	233.6	231	231.8
30	20	110	226	230.4	192.6	225.2	222.2	224.9	225.8	225.9	224.3	223.4	220.8	222.3
40	5	50	241	201.9	192.2	268	266.3	267.3	267.7	268.3	268.3	268	267	266.9
40	5	60	242.6	219.6	177.2	261.4	259.2	260.4	260.6	261.5	261.5	261.4	260.6	260.5
40	5	70	234	218.9	159.9	255.3	253.2	253.9	253.9	255.1	255.4	255.4	254.6	254.4
40	5	80	231.8	221.3	160.1	248.9	245.8	247.3	247.8	249.1	249.5	249.1	247.6	247.3
40	5	90	233.8	232.8	166.5	240.9	237.4	239.2	239.6	241.2	241.4	241.1	239.2	238.9
40	5	100	230.8	232.5	170	236.4	233	235.2	235.9	236.6	235.9	235.6	233.4	234.1
40	5	110	233.7	239.7	195.1	229.1	225.1	227.5	228.1	229.3	228.7	228.3	225.8	226.4
40	10	50	237.9	207.9	181.6	266.2	266	266.1	265.5	265.5	264.6	265.2	264.9	265.9
40	10	60	241.4	218	174.7	260.4	258.6	259.6	259.8	260.1	260.1	260.1	259.2	259.2
40	10	70	239.1	221	164.9	254.4	252.5	254	254.9	254.7	254.2	253.8	252.6	253
40	10	80	241.6	232.5	163.2	248.6	247.9	248.9	249	248.5	246.9	246.7	245.4	247.2
40	10	90	228	225	156.7	242.4	238.9	240.9	241.3	242.7	242.8	242.4	240.6	240.5
40	10	100	235.1	237.4	174.7	236.2	233.9	236.4	236.5	236.4	234.6	234.2	232	233.9
40	10	110	231.6	236.9	193.1	228.7	223.9	226.3	227.1	229	229.2	228.8	226.4	226.2
40	15	50	238.3	206.7	180.6	265.9	265.4	265.8	265.6	265.6	265	265.2	264.7	265.4
40	15	60	236.2	212.6	171	261.5	260.9	261.2	260.9	260.9	260.1	260.6	260.2	261
40	15	70	244.8	226.6	169.4	255.9	255.6	255.4	254.5	254.8	254.2	255.2	255.4	256
40	15	80	228.8	216.3	151.2	250.2	249.9	249.4	248.1	248.5	247.6	249.4	249.8	250.5
40	15	90	242.6	232.4	170.8	244.4	241.4	243.4	244.1	244.8	244.1	243.5	241.5	242.1
40	15	100	234.5	234	168.1	241.4	238.7	240.4	240.6	241.4	240.5	240.4	238.7	239.6
40	15	110	230.8	234.5	179.7	232.5	231.1	232.8	232.6	232.1	229.4	229.5	227.5	230.5
40	20	50	245.9	210.4	190.4	267.6	266.8	267.3	267.1	267.4	266.7	266.9	266.5	267
40	20	60	243	213.7	177.7	262.4	261.6	262.4	262.3	262	260.9	261.3	260.6	261.6
40	20	70	235.1	228.2	156.2	243.9	241.6	243.5	243.7	244.6	243.9	243.2	241.4	242
40	20	80	241.3	234.1	174.7	239.4	239.3	239.7	238.9	238.5	236.7	237.6	237	238.8
40	20	90	241.1	240.6	184	233.4	233.1	234.3	233.9	233.1	230.9	230.9	229.6	232.4
40	20	100	237.6	239.4	205.1	225.2	225.6	224.1	220.9	222	221.3	224.3	225.8	226.6
40	20	110	231.4	244.1	218.3	216.3	216	217.3	216.5	215.6	212.8	213.2	211.7	215